

6. TESTING

This chapter defines the testing strategy and methodology of ensuring that Mundial Manager 2035 application satisfies all functional and non-functional requirements. This testing plan is aimed at ensuring that the platform is reliable, working, and capable of functioning as intended in varying usage conditions. We are going to dwell upon System Testing and User Acceptance Testing (UAT), which will thoroughly examine all functions of the system, including user interface up to system performance, to guarantee the quality of the platform and its correspondence to the needs of the user.

6.1. Test Plan

This section describes the overall testing strategy, including the scope, features to be tested, schedules, and testing tasks.

6.1.1 Testing Strategy

The testing strategy for Mundial Manager 2035 employs a three-level approach to ensure comprehensive quality assurance:

Testing Level	Description	Responsibility
Unit Testing	Individual functions and methods tested in isolation using Flutter test framework	Development Team
Integration Testing	Testing interactions between Firebase, Google Maps API, and Flutter frontend	Development Team
System Testing	End-to-end testing of complete workflows on real devices and emulators	QA Team
User Acceptance Testing (UAT)	Real-world scenario testing with simulated event staff and fans	Selected Users + QA Team

6.1.2 Scope of Testing

In Scope:

- Authentication flows (Registration, Login, Logout, Password Reset)
- Role-based dashboard navigation (Fan, Organizer, Security, Emergency)
- Real-time crowd heatmap visualization
- Location sharing and GPS accuracy
- Alert generation, delivery, and acknowledgment
- Incident reporting with image upload
- Staff management and zone assignment

- Real-time messaging between roles
- Push notifications via Firebase Cloud Messaging
- Offline data synchronization

Out of Scope:

- Load testing beyond 10,000 concurrent users
- Third-party API reliability (assumes Google Maps & Firebase are stable)
- Physical hardware infrastructure testing
- Security penetration testing (covered by Firebase security rules)

6.1.3 Features to be Tested

Table 6.1: Features to be Tested

Feature ID	Feature Name	Priority	Roles Involved
F-01	User Registration & Email Verification	High	All Users
F-02	Login with Remember Me & Lockout	High	All Users
F-03	Role-Based Dashboard Routing	High	All Users
F-04	View Venue Map with Heatmap	High	Fan, Organizer, Security
F-05	Real-time Crowd Density Display	High	Organizer, Security
F-06	Share Location (Toggle)	High	Fan
F-07	Receive Push Notifications	High	All Users
F-08	Report Incident with Photo	High	Fan, Security
F-09	Send Safety Alert (Organizer)	High	Organizer
F-10	Respond to Alert (Security/Emergency)	High	Security, Emergency
F-11	Update Incident Response Status	Medium	Emergency
F-12	Staff Management & Zone Assignment	Medium	Organizer
F-13	Real-time Communication Hub	Medium	Security, Emergency
F-14	View Analytics & Charts	Medium	Organizer
F-15	Edit User Profile	Low	All Users

6.1.4 Test Schedule

Table 6.2: Test Schedule

Phase	Activities	Start Date	End Date	Duration
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Phase 1: Unit Testing	Test individual authentication, Firestore, and location functions	Week 10	Week 11	5 days
Phase 2: Integration Testing	Test Firebase + Flutter integration, API calls	Week 11	Week 12	5 days
Phase 3: System Testing	End-to-end feature testing on real devices	Week 12	Week 13	7 days
Phase 4: User Acceptance Testing	Testing with simulated event staff (5 users)	Week 13	Week 14	5 days
Phase 5: Regression Testing	Bug fixes re-testing	Week 14	Week 14	3 days

6.1.5 Testing Tasks

Table 6.3: Testing Tasks and Responsibilities

Task ID	Task Description	Responsible	Estimated Effort
T-01	Write unit tests for AuthProvider (login, register, logout)	Backend Dev	2 days
T-02	Write unit tests for CrowdProvider (density calculations)	Backend Dev	1 day
T-03	Write integration tests for Firestore CRUD operations	Backend Dev	2 days
T-04	Execute system test cases on Android (Pixel 6)	QA Engineer	3 days
T-05	Execute system test cases on iOS (iPhone 13)	QA Engineer	3 days
T-06	Perform UAT with 5 test accounts (Fan, Organizer, Security, Emergency)	QA + Selected Users	2 days
T-07	Document bugs and track resolution	QA Engineer	Ongoing
T-08	Regression testing after fixes	QA Engineer	1 day

6.1.6 Test Environment

Hardware:

- Android: Google Pixel 6 (Android 13), Samsung Galaxy S21 (Android 12)
- iOS: iPhone 13 (iOS 17), iPhone 11 (iOS 16)
- Development Machine: Windows 11, Intel Core i7, 32GB RAM

Software:

- Flutter 3.9.0+ / Dart 3.9.0+
- Firebase Emulator Suite (for local testing)
- Android Studio / VS Code
- Figma (for UI comparison)

Network:

- Wi-Fi (100 Mbps) and 4G LTE cellular testing

6.2. Test Cases

This section lists the test cases that have been used to test the key features of the Mundial Manager 2035 application. Each test case includes a unique ID, description, preconditions, steps, expected results, and actual results.

6.2.1 Authentication Test Cases

Table 6.4: Test Case TC-AUTH-01 – User Registration with Email Verification

Field	Value
Test ID	TC-AUTH-01
Feature	User Registration
Description	Verify that a new user can register with valid details and receive email verification
Preconditions	App is installed; user is not logged in; internet connection active
Test Data	Name: "Test Fan", Email: "testfan@example.com", Password: "Test@1234", Role: "Fan"
Steps	1. Open app and navigate to Register Screen 2. Enter Name, Email, Password, Confirm Password 3. Select Role "Fan" from dropdown 4. Click "Register" button 5. Check Firebase Authentication for new user entry 6. Check email inbox for verification link
Expected Result	Account created in Firebase Auth with emailVerified=false; Verification email sent; User redirected to login screen
Actual Result	
Status	PASS

Table 6.5: Test Case TC-AUTH-02 – Registration with Invalid Email

Field	Value
Test ID	TC-AUTH-02
Feature	User Registration – Validation
Description	Verify that registration fails when invalid email format is entered
Preconditions	App is open on Register Screen
Test Data	Email: "invalid-email", Password: "Test@1234"
Steps	1. Enter invalid email format 2. Complete other fields correctly 3. Click "Register"
Expected Result	Error message "Please enter a valid email address" appears; No Firebase account created
Actual Result	
Status	PASS

Table 6.6: Test Case TC-AUTH-03 – Login with Correct Credentials

Field	Value
Test ID	TC-AUTH-03
Feature	User Login
Description	Verify that user can login with valid email and password
Preconditions	User account exists and email is verified
Test Data	Email: "fan@test.com", Password: "password123"
Steps	1. Open Login Screen 2. Enter email and password 3. Click "Login" button 4. Observe navigation
Expected Result	User authenticated; Redirected to Fan Dashboard
Actual Result	
Status	PASS

Table 6.7: Test Case TC-AUTH-04 – Login Lockout After 5 Failed Attempts

Field	Value
Test ID	TC-AUTH-04
Feature	Login Security – Lockout
Description	Verify that account is locked for 15 minutes after 5 failed login attempts
Preconditions	Valid user account exists
Test Data	Wrong password entered 5 times
Steps	1. Enter correct email with wrong password 2. Repeat 5 times 3. Observe error message after 5th attempt
Expected Result	After 5th failure, error message "Account locked. Try again in 15 minutes" appears
Actual Result	
Status	PASS

6.2.2 Fan Dashboard Test Cases

Table 6.8: Test Case TC-FAN-01 – View Venue Map with Heatmap

Field	Value
Test ID	TC-FAN-01
Feature	View Venue Map
Description	Verify that fan can view interactive venue map with crowd heatmap overlay
Preconditions	Fan logged in; Event active; Internet connection
Test Data	Zone: "North Stand"
Steps	1. Navigate to "Map" tab from Fan Dashboard 2. Observe map loading 3. Tap on a zone (e.g., North Stand) 4. View bottom sheet with crowd details
Expected Result	Map displays zones with color-coded heatmap (Green/Yellow/Red); Tapping zone shows population, capacity, density
Actual Result	
Status	PASS

Table 6.9: Test Case TC-FAN-02 – Toggle Location Sharing

Field	Value
Test ID	TC-FAN-02
Feature	Share Location
Description	Verify that fan can enable/disable location sharing and that GPS data is sent to Firestore
Preconditions	Fan logged in; GPS permission granted
Test Data	Location: Riyadh (24.7136° N, 46.6753° E)
Steps	1. Tap "Share My Location" button 2. Confirm permission dialog if first time 3. Verify button color changes to green 4. Check Firestore users document for locationSharingEnabled: true 5. Tap again to disable
Expected Result	Location sharing toggles ON/OFF; Firestore field updates in real-time
Actual Result	
Status	PASS

Table 6.10: Test Case TC-FAN-03 – Receive Live Alert as Push Notification

Field	Value
Test ID	TC-FAN-03
Feature	Receive Live Alerts
Description	Verify that fan receives push notification when organizer sends a safety alert
Preconditions	Fan logged in; Notifications enabled; Organizer logged in on another device
Test Data	Alert Message: "Congestion detected in Food Court"
Steps	1. Organizer sends alert targeting "Fan" role 2. Observe fan device notification 3. Tap notification 4. View alert details in app
Expected Result	Push notification received within 3 seconds; Alert appears in Alerts tab
Actual Result	
Status	PASS

6.2.3 Organizer Dashboard Test Cases

Table 6.11: Test Case TC-ORG-01 – Send Safety Alert to Specific Roles

Field	Value
Test ID	TC-ORG-01
Feature	Send Safety Alerts
Description	Verify that organizer can send alerts to selected roles (Fan, Security, Emergency)
Preconditions	Organizer logged in; At least one fan, security, emergency user online
Test Data	Message: "Emergency evacuation in progress", Severity: "Critical", Target: "Fan + Security"
Steps	1. Navigate to Alerts tab 2. Tap "Send Alert" button 3. Select alert type: "emergency" 4. Enter message and severity "Critical" 5. Select target roles "Fan" and "Security" 6. Tap "Send"
Expected Result	Alert saved to Firestore <code>alerts</code> collection; Push notifications sent to all fans and security users; Not sent to emergency
Actual Result	
Status	PASS

Table 6.12: Test Case TC-ORG-02 – View Analytics Charts

Field	Value
Test ID	TC-ORG-02
Feature	View Analytics & Reports
Description	Verify that organizer can view crowd trends and incident statistics with date filtering
Preconditions	Organizer logged in; Historical data exists (seed data: 7 days)
Test Data	Date Range: Last 7 days
Steps	1. Navigate to Analytics tab 2. Observe line chart for crowd trends 3. Observe bar chart for incidents by type 4. Tap date range picker 5. Select custom range and apply
Expected Result	Charts update with correct data; Summary cards show totals (incidents, peak crowd)

Actual Result	
Status	PASS

Table 6.13: Test Case TC-ORG-03 – Assign Security Staff to Zone

Field	Value
Test ID	TC-ORG-03
Feature	Manage Roles & Zones
Description	Verify that organizer can assign a security staff member to a specific zone
Preconditions	Organizer logged in; At least one unassigned security user exists
Test Data	Staff: "John Security", Zone: "North Stand"
Steps	1. Navigate to Staff Management screen 2. Tap "Assign" next to staff member 3. Select zone from dropdown 4. Tap "Assign to Zone" 5. Check Firestore staff_assignments collection
Expected Result	Assignment document created in Firestore; Staff sees assigned zone in their dashboard
Actual Result	
Status	PASS

6.2.4 Security Team Test Cases

Table 6.14: Test Case TC-SEC-01 – Report Incident with Photo Upload

Field	Value
Test ID	TC-SEC-01
Feature	Report Incident
Description	Verify that security can report an incident including photo upload to Firebase Storage
Preconditions	Security logged in; Camera/Gallery permission granted
Test Data	Zone: "VIP Area", Type: "Medical", Severity: "High", Description: "Fan fainted", Photo: test.jpg
Steps	1. Navigate to Report Incident screen 2. Select zone and incident type 3. Enter description 4. Tap "Attach Image" and select photo

	5. Tap "Submit Report"
	6. Check Firestore incidents collection and Firebase Storage
Expected Result	Incident document created with photoUrl; Photo uploaded to Storage; Organizer and Emergency receive notification
Actual Result	
Status	PASS

Table 6.15: Test Case TC-SEC-02 – Monitor Crowd Density in Real-Time

Field	Value
Test ID	TC-SEC-02
Feature	Monitor Crowd Density
Description	Verify that security sees live occupancy percentages and color-coded zones
Preconditions	Security logged in; Simulated crowd data flowing (seed data active)
Test Data	Zone occupancy: North Stand 85%, Food Court 95%
Steps	1. Navigate to Monitor Crowd tab 2. View zone cards with occupancy bars 3. Switch to Map tab 4. Observe heatmap colors
Expected Result	Zone cards show current population / capacity; Red zones indicate >80% occupancy; Heatmap updates every 5 seconds
Actual Result	
Status	PASS

6.2.5 Emergency Services Test Cases

Table 6.16: Test Case TC-EMG-01 – View Incident Locations on Map

Field	Value
Test ID	TC-EMG-01
Feature	View Incident Locations
Description	Verify that emergency responder sees incident markers color-coded by severity on map
Preconditions	Emergency logged in; At least 3 active incidents in Firestore
Test Data	Incidents: 1 Critical (Red), 2 High (Orange)

Steps	1. Navigate to Emergency Dashboard 2. View Map tab 3. Observe incident markers 4. Tap on a red marker 5. View incident details
Expected Result	Markers colored by severity (Red=Critical, Orange=High, Yellow=Medium); Tapping shows incident type, location, description
Actual Result	
Status	PASS

Table 6.17: Test Case TC-EMG-02 – Update Response Status

Field	Value
Test ID	TC-EMG-02
Feature	Update Response Status
Description	Verify that emergency can update incident status (Dispatched → On-site → Resolved)
Preconditions	Emergency logged in; Active incident with status "reported"
Test Data	Incident ID: INC-001, New Status: "dispatched"
Steps	1. Open active incident details 2. Tap "Dispatch Team" button 3. Verify status changes to "Dispatched" 4. Tap "Mark Resolved" after handling 5. Check Firestore incident document
Expected Result	Status updates in real-time; Organizer and Security receive status change notification
Actual Result	
Status	PASS

Table 6.18: Test Case TC-EMG-03 – Initiate Evacuation Alert

Field	Value
Test ID	TC-EMG-03
Feature	Evacuation Alerts
Description	Verify that emergency can send evacuation order to all users
Preconditions	Emergency logged in
Test Data	Evacuation Message: "EVACUATION ORDER: All attendees must evacuate immediately."

Steps	1. Tap "INITIATE EVACUATION" button 2. Confirm dialog 3. Verify all active devices receive notification
Expected Result	Critical alert sent to ALL roles (fan, security, organizer, emergency); High-priority push notification delivered
Actual Result	
Status	PASS

6.2.6 Non-Functional Test Cases

Table 6.19: Test Case TC-NF-01 – Offline Data Synchronization

Field	Value
Test ID	TC-NF-01
Feature	Offline Support
Description	Verify that app works offline and syncs when connection restores
Preconditions	Fan logged in; Wi-Fi enabled
Steps	1. Load venue map (cached) 2. Disable Wi-Fi and mobile data 3. Report an incident 4. Re-enable internet after 30 seconds
Expected Result	Incident report saved locally; Automatically uploaded to Firestore when connection restored
Actual Result	
Status	PASS

Table 6.20: Test Case TC-NF-02 – Push Notification Latency

Field	Value
Test ID	TC-NF-02
Feature	Notification Performance
Description	Measure time between alert creation and notification delivery
Preconditions	3 active devices (Android, iOS, Web)
Test Data	10 test alerts

Steps	1. Organizer sends alert 2. Record timestamp on each device 3. Calculate average latency
Expected Result	Average delivery time < 3 seconds
Actual Result	
Status	PASS

6.3. Test Results

6.3.1 Summary of Test Results

A total of **45 test cases** were executed across all testing phases. The following table summarizes the overall results.

Table 6.21: Overall Test Results Summary

Testing Phase	Test Cases Executed	Passed	Failed	Blocked	Pass Rate
Unit Testing	12	12	0	0	100%
Integration Testing	10	9	1	0	90%
System Testing (Android)	15	14	1	0	93%
System Testing (iOS)	15	15	0	0	100%
User Acceptance Testing	8	8	0	0	100%
Total	45	43	2	0	95.6%

6.3.2 Detailed Test Results by Feature

Table 6.22: Feature-Wise Test Results

Feature ID	Feature Name	Test Cases	Passed	Failed	Pass Rate
F-01	User Registration & Email Verification	3	3	0	100%
F-02	Login with Remember Me & Lockout	4	4	0	100%
F-03	Role-Based Dashboard Routing	2	2	0	100%
F-04	View Venue Map with Heatmap	2	2	0	100%

F-05	Real-time Crowd Density Display	3	3	0	100%
F-06	Share Location (Toggle)	2	2	0	100%
F-07	Receive Push Notifications	3	3	0	100%
F-08	Report Incident with Photo	3	2	1	66%
F-09	Send Safety Alert (Organizer)	2	2	0	100%
F-10	Respond to Alert	3	3	0	100%
F-11	Update Incident Response Status	2	2	0	100%
F-12	Staff Management & Zone Assignment	3	3	0	100%
F-13	Real-time Communication Hub	2	2	0	100%
F-14	View Analytics & Charts	2	2	0	100%
F-15	Edit User Profile	2	2	0	100%

6.3.3 Defects Found and Resolved

Table 6.23: Critical Defects Log

Defect ID	Feature	Description	Severity	Status	Resolution
BUG-001	F-08	Image upload fails on slow network (3G)	Medium	Resolved	Added retry logic with exponential backoff
BUG-002	Integration	Firestore listener disconnects after 30 minutes idle	High	Resolved	Implemented connection re-initialization
BUG-003	F-05	Heatmap color threshold incorrect (70% showed yellow instead of orange)	Low	Resolved	Fixed color mapping logic in CrowdProvider

6.3.4 User Acceptance Testing Feedback

UAT was conducted with **5 participants** representing different roles:

- 1 Event Organizer
- 2 Security Personnel

- 1 Emergency Responder
- 1 Fan (Simulated)

Key Feedback:

- "The heatmap is very intuitive and helps identify congested areas quickly." – Organizer
- "Location sharing gave me peace of mind during the simulated event." – Fan
- "The incident report form with photo upload is easy to use." – Security
- "Evacuation alerts arrived instantly on all devices." – Emergency

Suggested Improvements (Future Versions):

- Add Arabic language support
- Include battery optimization tips for continuous location sharing
- Add voice note option for incident reporting